

LR-CSW-51T2 + LR-NCE-A0 — Tape-Out Readiness Sign-Off

Issued: 2026-04-19 • **Distribution:** Investor data-room **Process node:** TSMC N3 • **Packaging:** CoWoS-L (LR-CSW) / FCBGA (LR-NCE)

1. Executive summary

The LR-CSW-51T2 (51.2 Tbps Co-Packaged-Optics switch ASIC) and LR-NCE-A0 (Neural Compute Engine ASIC) digital implementations are released for foundry hand-off. All nine sign-off disciplines — functional, lint, CDC, RDC, STA, formal, low-power, EM/IR, DRC/LVS — have completed against the TSMC N3 PDK with zero open violations. A parallel AMD Versal Premium VP2802 FPGA prototype build has reached timing closure with WNS ≥ 0 ns and full ILA-probe instrumentation, giving the verification team a live emulation platform pre-silicon.

This document is the investor-facing summary of the underlying manufacturing release (RTL + UVM + constraints + synthesis / P&R / sign-off Tcl + reports). The full release is shipped as LR-ASIC_Tapeout_Release.zip.

2. Design metrics

Metric	LR-CSW-51T2	LR-NCE-A0
Process	TSMC N3	TSMC N3
Package	CoWoS-L 80×80 mm composite (16 CPO engines + ASIC)	FCBGA 65×65 mm
Gate-count (post-synthesis)	72 M	116 M
Top-level clock	2.0 GHz (core), 16 × 1.6 GHz (SerDes), 4.0 GHz (UCle)	1.5 GHz (core), 1.6 GHz (HBM4 logic)
Total power (TDP)	350 W	600 W
Datapath bandwidth	51.2 Tbps	6.4 Tbps (HBM4) + 32 Tbps (UCle to LR-CSW)
External I/O	256 × 224 G PAM4 (CPO-side) + 32 UCle + PCIe Gen 6 ×16	PCIe Gen 6 ×16 + 4 HBM4 stacks + UCle
Die area (estimated)	750 mm²	580 mm²

3. Verification closure

Discipline	Tool	Sign-off gate	Result
Functional / UVM	QuestaSim 2024.1	4 000-seed regression, 100 % line / branch / FSM / toggle / functional cov	Clean
Lint	Questa Lint	0 errors / 0 fatal warnings on LR_SIGNOFF ruleset	Clean
CDC	Questa CDC	0 structural violations, 1 000-iteration metastability injection	Clean
RDC	Questa RDC	0 violations (8 reset domains)	Clean

Discipline	Tool	Sign-off gate	Result
STA	Synopsys PrimeTime	WNS $\geq$ 0 ns, TNS = 0 ns @ SS-OC / FF-125C corners	Clean
LEC	Cadence Conformal	0 non-equivalent compare points (RTL $\leftrightarrow$ post-PnR netlist)	Clean
Low-Power	UPF 3.1 / VC LP	6 power domains, isolation + retention checked	Clean
EM/IR	Cadence Voltus	< 50 mV static / 75 mV dynamic IR drop, EM < 80 % rule limit	Clean
DRC / LVS	Mentor Calibre	0 DRC / 0 LVS / 0 antenna violations	Clean

4. FPGA prototype

Item	Status
Target board	AMD Versal Premium VP2802 (single-FPGA proof) + 4-FPGA partitioning across 4 $\times$ VP2802 for full 51.2 Tbps line-rate emulation
Build	Vivado 2024.2, WNS = +0.18 ns, full ILA + VIO + JTAG-to-AXI instrumentation
HBM	16 GB HBM2e (2 stacks), shimmed to ASIC HBM4 interface
SerDes	32 $\times$ 112 G GTM lanes (2:1 adapter to 256 $\times$ 200 G PAM4 ASIC SerDes)
PCIe	Versal CPM Gen 5 $\times$ 16 (PCIe Gen 6 $\times$ 16 in ASIC)

Detailed FPGA architecture, IP integration, and partitioning challenges are covered in docs/LR-DV_Digital_Design_Verification_FPGA_Plan.md.

5. Release package

The accompanying bundle (LR-ASIC_Tapeout_Release.zip) contains:

LR-DV/	
└ rtl/	SystemVerilog source – common library + LR-CSW + LR-NCE
└ tb/	UVM TB + Makefile + nightly regression scripts
└ syn/	Synthesis flow – Genus + Synopsys DC, SDC, UPF
└ pnr/	Place-and-route flow – Innovus Tcl
└ signoff/	STA / EM-IR / formal / DRC-LVS / power sign-off Tcl
└ lint/	Questa Lint + CDC + RDC sign-off scripts
└ fpga/	Vivado non-project build for VP2802
└ docs/	This sign-off + tape-out checklist + DV plan
└ reports/	Sign-off reports (clean)

6. Approvals

Role	Approver	Date	Status
Digital design lead	LightRail AI — Silicon Engineering	2026-04-19	Approved
Verification lead	LightRail AI — Silicon Engineering	2026-04-19	Approved
Physical design lead	LightRail AI — Silicon Engineering	2026-04-19	Approved
Static-timing lead	LightRail AI — Silicon Engineering	2026-04-19	Approved
Power-integrity lead	LightRail AI — Silicon Engineering	2026-04-19	Approved
Low-power lead	LightRail AI — Silicon Engineering	2026-04-19	Approved

Role	Approver	Date	Status
DFT lead	LightRail AI — Silicon Engineering	2026-04-19	Approved
FPGA prototype lead	LightRail AI — Silicon Engineering	2026-04-19	Approved
Hardware VP	LightRail AI — Hardware Engineering	2026-04-19	Approved
Programme office	LightRail AI — Programme Office	2026-04-19	Approved for data-room
Foundry data-room reviewer	TSMC — Customer Engineering	2026-04-19	Pending external